

Diffraction minimum relationship

In a diffraction experiment, the minimum of order one is subtended at an angle $\theta = 30^\circ$. It can be deduced that:

- a. the width of the slit is double the wavelength
- b. the distance between the slit and the screen is half the wavelength
- c. the distance between the slit and the screen is double the wavelength
- d. the width of the slit is half the wavelength

Relation between group and phase speed

The relation between the group speed v_g and the phase speed v_f is

a. $v_g = v_f + \lambda dv_f/d\lambda$

b. $v_g = v_f - dv_f/d\lambda$

c. $v_g = v_f + dv_f/d\lambda$

d. $v_g = v_f - \lambda dv_f/d\lambda$

Definition of absolute refractive index

The absolute refractive index of a substance indicates:

- a. the ratio between the propagation speed of an electromagnetic wave in the material and that of the same wave in vacuum
- b.** the ratio between the propagation speed of an electromagnetic wave in vacuum and that of the same wave in the material
- c. the inverse of the angle with which an electromagnetic wave coming from the vacuum affects the material.
- d. the inverse of the propagation speed of an electromagnetic wave in the material.

Solution of the wave equation

The function $\Psi(x, t) = A \cdot \exp[-(kx - \omega t)]$ can be the solution of the wave equation?

- a. It depends from the value of the amplitude A.
- b. It depends from the value of the wave number k
- ☒ c. Yes
- d. No.

Gaussian pulse and the wave equation

Given the expression $f(x, t) = A \cdot \exp[-b(x - vt)^2]$, where A and b are positive real constants given in appropriate units measurement. Which of the following statements is correct?

- a. $f(x, t)$ satisfies the wave equation only for $b \ll 1$.
- b. $f(x, t)$ does not satisfy the wave equation.
- c. $f(x, t)$ cannot satisfy the wave equation because the dependence on $(x - vt)$ is quadratic instead of linear.
- ☒ d. $f(x, t)$ satisfies the wave equation.

Physical meaning of Poynting vector flux

The flow of the Poynting vector through a closed surface S represents

- ☒ a. the variation over time of the electromagnetic energy contained in the volume V delimited by the closed surface S
- ☐ b. the energy dissipated by the Joule effect in the volume V delimited by the closed surface S
- ☐ c. the electromagnetic energy contained in the volume V bounded by the closed surface S
- ☐ d. the amplitude of the electromagnetic wave that crosses the closed surface S

Width of slit in diffraction

A monochromatic light beam ($\lambda = 644 \text{ nm}$) passes through a rectangular slit and forms a figure of diffraction in which the first dark fringe corresponds to $\theta = 0.125^\circ$. What is the width of the slit?

- a. $2.95 \cdot 10^{-4} \text{ m}$
- b. $1.66 \cdot 10^{-4} \text{ m}$
- c. $3.8 \cdot 10^{-4} \text{ m}$
- d. $1.25 \cdot 10^{-4} \text{ m}$

Intensity of a cylindrical wave

The intensity of a cylindrical wave at a distance r from the source

- ☒ a. decreases as $1/r$
- ☐ b. doesn't change
- ☐ c. decreases as $1/\sqrt{r}$
- ☐ d. decreases as $1/r^2$

Wavelength and period of EM wave

An electromagnetic wave having a frequency of $1.8 \cdot 10^{12}$ Hz propagates at the speed $v = 2.7 \cdot 10^8$ m/s. Determine the value of the wavelength λ , and of the period T .

a. $\lambda = 1.5 \cdot 10^{-2}$ m, $T = 5.55 \cdot 10^{-12}$ s

b. $\lambda = 1.5 \cdot 10^{-4}$ m, $T = 5.55 \cdot 10^{-12}$ s

c. $\lambda = 1.5 \cdot 10^{-3}$ m, $T = 5.55 \cdot 10^{-13}$ s

d. $\lambda = 1.5 \cdot 10^{-4}$ m, $T = 5.55 \cdot 10^{-13}$ s

Intensity of EM wave in dielectric medium

A plane harmonic electromagnetic wave is described by the following relationship: $E(x, t) = E_0 \sin(kx - \omega t)$ and propagates in a medium having relative dielectric constant. In the hypothesis that the medium is diamagnetic and the refractive index is n , the intensity I of the wave is

a. $I = \epsilon_0 \epsilon_r c E_0^2 \sin^2(kx - \omega t) / n$

b. $I = \epsilon_0 \epsilon_r c E_0^2 \sin^2(kx - \omega t) / (2n)$

c. $I = \epsilon_0 \epsilon_r c E_0^2 / (2n)$

d. $I = \epsilon_0 c E_0^2 / (2n)$

Conditions for interference

In order to observe the phenomenon of interference between the waves emitted between two coherent sources, it is necessary that

- a. the two waves propagate in the same medium.
- b. the difference of path that the two waves must travel to reach a point on the screen must be much greater than their length
- ☒ c. the difference of path that the two waves must travel to reach a point on the screen must be much less than their length
- d. the two waves propagate in different media

The wave equation

The wave equation for a function $\Psi(x, y, z, t)$ is

a.
$$\frac{\partial^2 \Psi}{\partial x^2} + \frac{\partial^2 \Psi}{\partial y^2} + \frac{\partial^2 \Psi}{\partial z^2} = \frac{1}{v} \frac{\partial \Psi}{\partial t}$$

b.
$$\frac{\partial^2 \Psi}{\partial x^2} + \frac{\partial^2 \Psi}{\partial y^2} + \frac{\partial^2 \Psi}{\partial z^2} = \frac{1}{v} \frac{\partial^2 \Psi}{\partial t^2}$$

c.
$$\frac{\partial \Psi}{\partial x} + \frac{\partial \Psi}{\partial y} + \frac{\partial \Psi}{\partial z} = \frac{1}{v} \frac{\partial \Psi}{\partial t}$$

d.
$$\frac{\partial^2 \Psi}{\partial x^2} + \frac{\partial^2 \Psi}{\partial y^2} + \frac{\partial^2 \Psi}{\partial z^2} = \frac{1}{v^2} \frac{\partial^2 \Psi}{\partial t^2}$$